

Scalability & Future Plan

Date	30 September 2026
Team ID	SWTID-2026-1307
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The field and value (Priority = 1) are fixed inside the ACL script.	Any change in the rule needs the script to be edited.	Medium
2	The ACL script runs for every record in a list.	Very large lists may load slowly.	High
3	Only the Read operation of the Incident table is protected.	Write, delete and other tables are not covered.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Tested with two users (Manager User and Normal User) on a developer instance.	Assign access through roles and groups so that any number of users is covered without changing the ACL.
2	Data Storage	Sample incidents stored in the Incident table of the PDI.	Deploy on a production instance, archive old incidents and add an index on the controlling field.
3	Performance	Simple script with a single field check and role check.	Keep the script light, move logic to a Script Include and filter records with a Before Query Business Rule.
4	Security	Record-level Read ACL based on field value and role.	Add write, delete and field-level ACLs, log denied access attempts and test regularly with impersonation.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Read ACL with script for Priority 1 records, testing and Update Set deployment.	Completed (Sep 2026)	Sensitive incidents are protected from normal users.
Phase 2	Extend the ACL to write and delete operations and add field-level ACLs.	Q4 2026	Complete protection for viewing and editing records.
Phase 3	Make the restricted field and value configurable (system property or table) and use group-based rules.	Q1 2027	Rules can be changed without editing the script.
Phase 4	Apply the solution to multiple tables and add an audit log dashboard for denied access.	Q2 2027	Organization-wide data security with better monitoring.